From ETF Allocation to Automated Trading

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Programming Assignment 3

1. From choosing the fund to running it

Earlier assignments defined the jobs of the portfolio, chose ETFs for the growth block, and studied how to divide the money among them. This assignment takes the next step: turning those choices into rules for buying, selling and rebalancing. I then test whether a trend rule helps the fund after costs.

The question moves from what to hold and how much to invest to how to maintain the portfolio over time. The safety and inflation blocks keep their original jobs. This study trades only the growth fund.

The question

Can a monthly trend rule reduce the fund's losses without giving up too much growth? The optimizer decides how much each ETF should receive. The trend rule decides whether that amount stays in the ETF or moves to cash.

How the rule works

Step	Decision
Choose eligible ETFs	Use the earlier catalogue. Require five years of returns and enough trading volume; remove equivalent wrappers.
Set the weights	Update the Checkpoint B allocation using only earlier data. Use no short positions or borrowing.
Check the trend	Compare each ETF's month-end USD price with its ten-month average. Keep its target above the average; move it to cash otherwise.
Trade later	Use the next available common closing date. Trade when weights leave their bands; charge purchases and sales.

All 61 catalogue ETFs remain candidates. Between 28 and 38 qualify at each of the 87 monthly decisions. A young fund must build enough history before it can enter.

Each ETF can receive at most 20% at a decision. The fund also has a model-based monthly loss limit. I keep three-percentage-point trading bands: small weight changes can wait, while an ETF rejected by the trend rule must be sold. Appendix A gives the exact rules.

The ten-month rule follows Faber [2]. A sustained decline may continue after a sell signal. A quick recovery can also leave the fund in cash while prices rise. That trade-off is what I test.

2. What happened to return and risk?

The test covers 2019-07-01 to 2026-09-30, in USD. I compare the fund with bands, the same fund with trend, and three buy-and-hold controls. The initial-fund control buys the first allocation once. SPY provides the required S&P 500 benchmark; VT adds a global-equity reference.

Account	CAGR	Volatility	Sharpe	Max DD
Initial fund, buy-and-hold	13.64%	18.75%	0.624	-33.93%
Fund with bands	12.10%	17.79%	0.570	-34.35%
Fund + trend	10.20%	11.06%	0.674	-16.37%
SPY buy-and-hold	15.60%	19.60%	0.692	-33.72%
VT buy-and-hold	12.85%	18.82%	0.585	-34.24%

CAGR is compound annual growth. Volatility measures return variation. Sharpe measures return above the cash proxy relative to its variation. Maximum drawdown is the largest loss from a previous wealth peak.

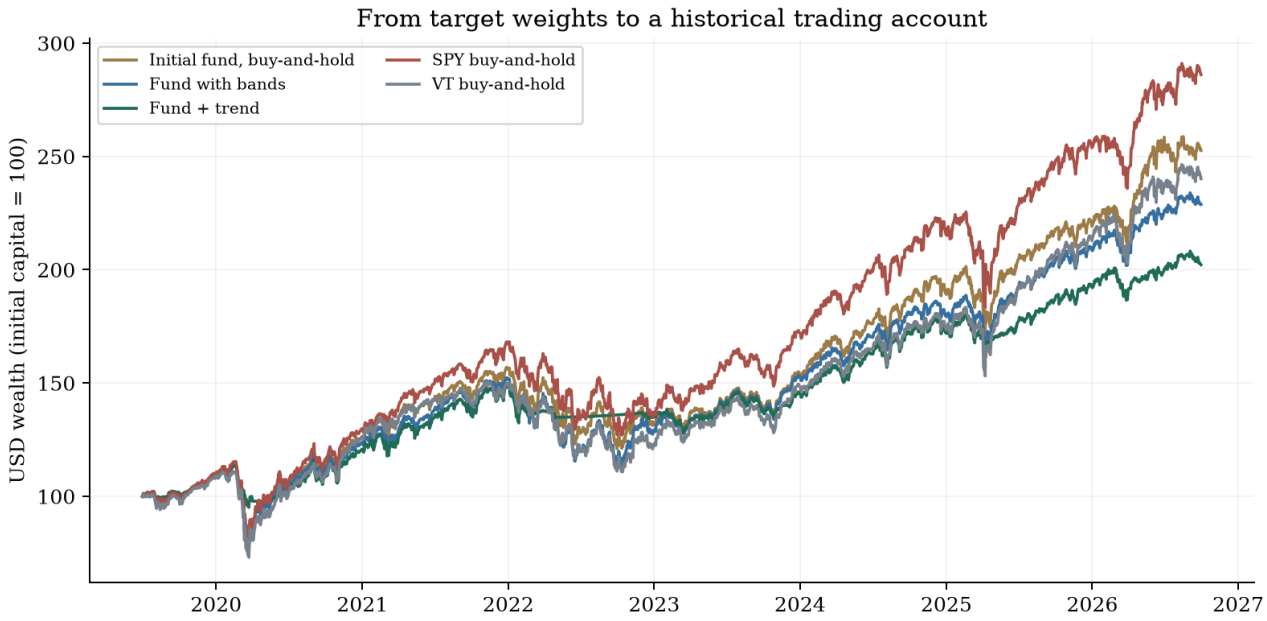


Figure 1. Net historical wealth, starting from 100. All accounts use the same main dates.

With trend, CAGR is 10.20%, versus 12.10% for the fund with bands. Sharpe is 0.674, versus 0.570. The filter changes both growth and risk; it does not simply add return to the original fund.

SPY has CAGR 15.60% and Sharpe 0.692. The fund therefore needs to be judged against a simple alternative, even when its own risk measures improve.

I assume a cost of 0.10% of every amount bought or sold. The three fund accounts also pay a 0.25% annual management expense. SPY and VT pay no extra fund-management fee. Cash earns a proxy based on three-month Treasury yields. These are modeling assumptions; Appendix A explains the accounting.

3. What does the lower drawdown mean?

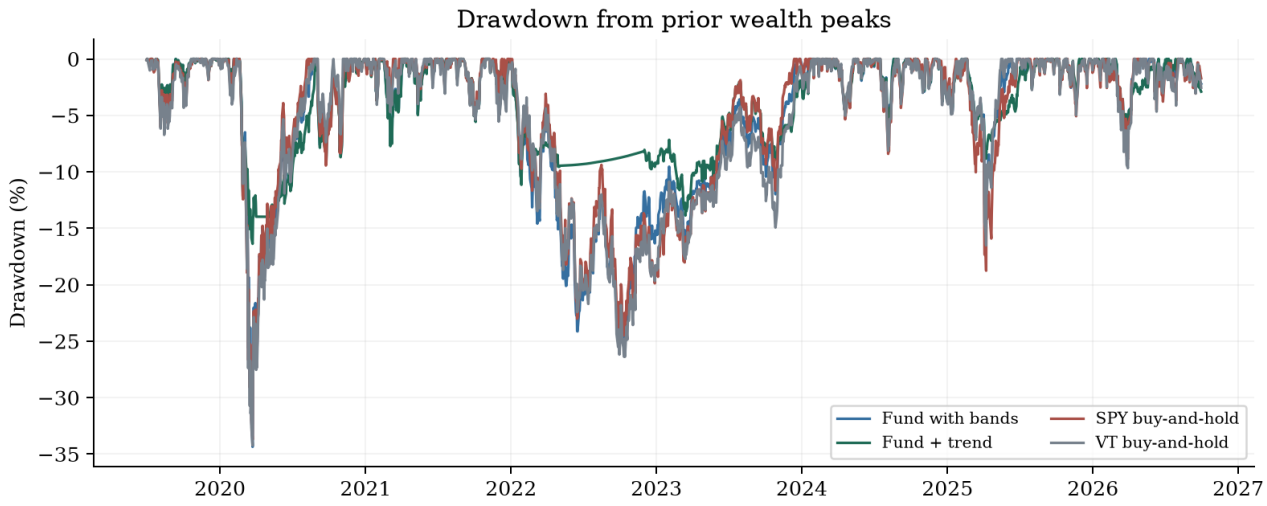


Figure 2. Loss from each account's previous wealth peak.

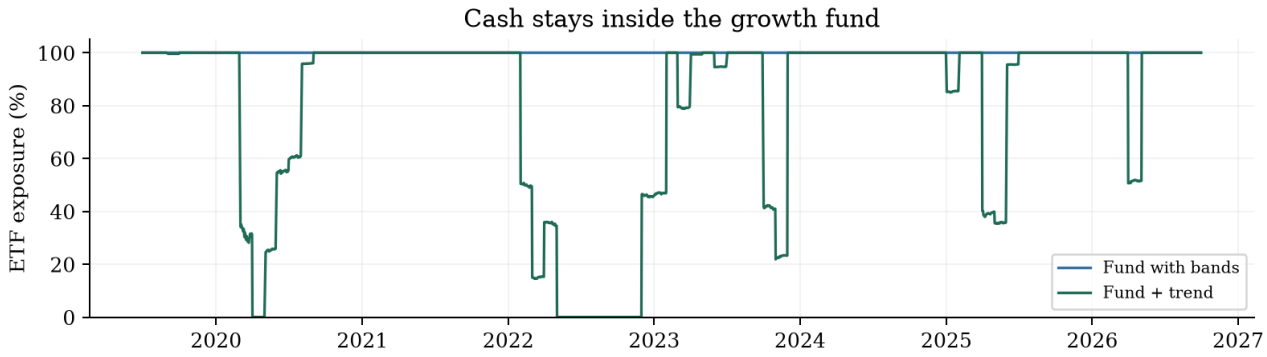


Figure 3. ETF exposure as a share of fund wealth. The remainder stays in growth-fund cash.

The largest fall is -16.37% with trend and -34.35% without it. In this history, leaving some allocations in cash reduced the depth of the loss.

The trend fund holds 80.34% in ETFs on average. That matters: part of the lower risk comes from being less invested. The cash belongs to the growth fund and does not replace the separate safety reserve.

The filter also trades more. Its annual turnover is 1.55 times fund wealth, compared with 0.69 for the fund with bands. Each purchase and sale has a cost.

For the investor, the choice is between more time in the market and a rule that may soften a prolonged decline. The filter can also exit late, miss a recovery or trade repeatedly around the moving average. Monthly decisions leave the fund exposed to losses between signals.

4. Is the improvement reliable enough to adopt?

I split the ongoing accounts into an earlier and a later period. Holdings carry across the boundary; the fund does not restart or pay a second entry cost.

Account	Period	CAGR	Sharpe	Max DD
Fund with bands	earlier	7.43%	0.384	-34.35%
Fund with bands	later	16.63%	1.069	-14.60%
Fund + trend	earlier	8.97%	0.649	-16.37%
Fund + trend	later	11.36%	0.739	-8.83%
SPY buy-and-hold	earlier	9.38%	0.457	-33.72%
SPY buy-and-hold	later	21.73%	1.086	-18.76%

Earlier means July 2019-December 2022; later means January 2023-September 2026. Previous work already examined overlapping dates. This split shows how the result changes over time; it is not a new, untouched test.

I also change trading costs, the trend window, the bands and the execution delay. I keep the ten-month rule as the main result, even if another setting looks better. Appendix B reports every alternative.

A block-resampling check asks how much the Sharpe difference varies when groups of observed months are drawn again. Its 95% interval is [-0.303, +0.425]. It includes zero, so this check does not establish a stable Sharpe advantage. The detailed assumptions are in Appendix B.

Conclusion: automate the process; keep trend experimental

Automated execution worked: the program turned the policy into delayed trades and accounts that reconcile. The added management rules did not show a clear investment advantage in this sample.

Rebalancing had CAGR 12.10% and Sharpe 0.570. Holding the initial allocation had 13.64% and 0.624. The extra allocation decisions did not improve growth or Sharpe.

Trend delivered protection instead: maximum drawdown changed from -34.35% to -16.37%, while CAGR fell from 12.10% to 10.20%. It softened the worst fall by holding more cash and trading more. That protection had a growth cost; the Sharpe check above does not establish a stable advantage.

My decision for the project is to keep the automated execution and accounting, retain buy-and-hold as the reference, and leave trend out of the default fund policy. The growth block is meant to compound. Adding a cash-exit rule requires deciding how much growth to sacrifice for a smaller drawdown and testing that choice further. This exercise has not justified that change.

This conclusion applies to the tested period and assumptions. The catalogue and data are retrospective, and historical company/sector limits remain unverified. It does not establish that buy-and-hold will always win.

Appendix A. Data and allocation

The inputs are the real observations saved in Module 6. Prices end October 23, 2026; the study stops at September 30, the last complete month. The normal run is offline.

Input	Use
daily_prices.csv	Adjusted local prices, including the vendor's split and distribution adjustments
fx.csv	Convert BRL, GBP and EUR prices to USD before calculating returns
daily_liquidity.csv	Rebuild eligibility from the prior six calendar months of trading volume
risk_free.csv	DGS3MO Treasury yield for the cash proxy and excess-return metrics
universe.csv	Fixed 61-ETF catalogue and equivalent-wrapper groups
Source/provenance JSON	Retrieval metadata, original source URLs and copy hashes

Eligibility requires 61 consecutive monthly USD prices, giving 60 returns. Median daily dollar volume must reach USD 1 million, with at least 100 valid observations in the prior six months. Wrapper preference follows the fixed order from Checkpoint B. It is applied among eligible funds, not by ranking future returns.

FX uses the same date or an earlier observation, no more than five calendar days old. Monthly prices use each ETF's last observation within seven days of month-end. Fourteen early PIBB11 volume records lack dollar volume; the raw records stay intact and the filter excludes them. Dividends are not added again to adjusted prices.

How the optimizer sets weights

The optimizer uses the prior 60 monthly returns. Ledoit-Wolf shrinkage stabilizes the covariance matrix [5]. Individual means move toward a common estimate, as in Checkpoint B. Here mu_0 is the minimum-variance portfolio's sample mean and lambda is a data-driven shrinkage weight. This reduces the influence of noisy estimates without making them reliable forecasts.

$$\hat{\mu} = 12[(1-\lambda)\bar{r} + \lambda\mu_0\mathbf{1}]$$

It seeks the highest estimated excess return per unit of volatility after forecast costs. The equation below states that objective. Annual management cost is m, one-way trading cost is c, and w_prev is the previous structural target.

$$\max_{w} \frac{\hat{\mu}^T w - r_f - m - 12c\|w - w_{\text{prev}}\|_1}{\sqrt{w^T\hat{\Sigma}w}}$$

Targets sum to one, prohibit shorts, cap each ETF at 20%, and limit empirical 95% monthly CVaR to 15%, including forecast costs. CVaR estimates average loss in the worst 5% of the training returns; it is not a guaranteed loss ceiling. CLARABEL solves the transformed conic problem. If estimated net excess opportunities are nonpositive, a recorded minimum-risk fallback applies.

Both accounts use the same structural targets and previous-target cost reference. Actual costs use each account's actual trades. Company, sector, country and economic-currency limits from the full Checkpoint B mandate are not imposed using future composition snapshots. Shorts and leverage are disabled in this operational extension.

Appendix A. Execution and measurement

The trend filter compares the current adjusted USD month-end price with its ten-month average, including the current month. A positive test retains the structural target; otherwise the target is zero. The structural targets of other ETFs stay unchanged.

$$w_{i,t}^{\text{trend}} = w_{i,t}^{\text{base}} \mathbf{1}\left[P_{i,t} > \tfrac{1}{10}\sum_{j=0}^{9} P_{i,t-j}\right]$$

Execution waits for the first later SPY-calendar date with actual quotes for potential holdings and controls. Each actual purchase or sale is checked again. A delay beyond ten calendar days fails. Valuation alone may carry an earlier observed USD price for up to seven days. Different market closes and carried valuations can affect daily risk estimates.

When weights leave their three-percentage-point bands or fail the risk check, a linear program minimizes turnover within the bands and ETF cap. Cash returns to its desired target when a trade occurs. A zero target forces full exit. Weights can drift between decisions, so decision-date limits are not continuous guarantees.

Costs reduce the capital available for the new holdings. The account solves the following identity, where H is the value held before the trade. Purchases and sales both count.

$$V_{\text{after}} + c\sum_{i}|w_i V_{\text{after}} - H_{i,\text{before}}| = V_{\text{before}}$$

Initial capital is one USD accounting unit. Fractional adjusted-price units and reinvested distributions are assumed. Fund management expense accrues ACT/365 proportionally against holdings and cash, without borrowing. Embedded ETF expenses are not deducted again. Cash uses the DGS3MO yield known strictly before each interval starts, times elapsed days / 365. No taxes, FX spreads, capacity or market impact are modeled. The final account is marked to market without a forced sale.

Metric	Calculation
CAGR	Compounded net wealth, annualized by elapsed calendar years
Volatility	Sample daily-return standard deviation times sqrt(252)
Sharpe	Mean daily excess return / its sample standard deviation, times sqrt(252); all accounts use the same cash proxy
Maximum drawdown	Largest loss from a running wealth peak, including pre-cost initial capital
Annual turnover	Absolute buys plus sells / pre-trade NAV, summed and annualized by elapsed years
Cost totals	Sum of dollar costs in original-capital units, not annual rates or shares of final NAV

The first row includes purchase costs with zero elapsed market return. Segment CAGR begins at the preceding observation. Buy-and-hold accounts trade once; the initial-fund control allows its weights to drift. SPY and VT pay initial trading costs but no additional fund-management expense.

Appendix B. Sensitivity and exposure checks

Alternative	Account	CAGR	Sharpe	Max DD
Cost 0 bp / side	Core	12.24%	0.570	-33.05%
Cost 0 bp / side	Trend	11.43%	0.694	-18.48%
Cost 10 bp / side	Core	12.10%	0.570	-34.35%
Cost 10 bp / side	Trend	10.20%	0.674	-16.37%
Cost 25 bp / side	Core	12.68%	0.602	-32.49%
Cost 25 bp / side	Trend	9.55%	0.589	-20.23%
Trend 6 months	Trend	9.63%	0.650	-13.61%
Trend 12 months	Trend	10.31%	0.667	-16.98%
Exact targets; no bands	Core	12.08%	0.570	-34.84%
Exact targets; no bands	Trend	10.07%	0.667	-16.43%
Second common execution day	Core	11.88%	0.560	-34.36%
Second common execution day	Trend	9.90%	0.647	-17.82%

Cost alternatives rerun the allocation objective and charge actual costs. They do not subtract fees from identical holdings. The delayed accounts start one common trading date later. The main rule remains ten months, ten basis points and three-percentage-point bands.

What the resampling check tells us

The stationary block bootstrap draws 2,000 paired paths from 87 observed monthly core, trend and cash returns, using mean blocks of six months and seed 4510708. The monthly-annualized Sharpe difference is +0.071; its percentile 95% interval is [-0.303, +0.425]. Monthly and daily Sharpe conventions differ.

The interval spans zero. This check does not establish a stable positive difference. It is conditional on the observed returns, does not rerun portfolio selection, and gives neither a null-hypothesis p-value nor a full correction for earlier research choices.

How much comes from holding cash?

A separate diagnostic mixes core-account returns with cash at the trend fund's full-sample mean ETF exposure. It gives CAGR 10.51%, Sharpe 0.570 and drawdown -28.33%. That weight is known only after observing the sample. The mix is an ex-post exposure check, not an investable or volatility-matched benchmark, and cannot isolate timing skill.

Appendix C. Trades, checks and reproduction

This excerpt shows the largest absolute trades from 2020-03-02, following the 2020-02-29 signal. Values are fractions of original capital; a negative amount is a sale. The complete ticket file retains every trade.

ETF	Side	USD notional	Cost	Adjusted USD price
USMV	SELL	-0.207944	0.000208	57.7812
DVY	SELL	-0.197597	0.000198	75.1768
DGRO	SELL	-0.181111	0.000181	33.5793
SPHQ	SELL	-0.134205	0.000134	31.7758
MTUM	BUY	+0.022802	0.000023	118.0440
VOOG	BUY	+0.014037	0.000014	27.4717
QQQ	SELL	-0.000088	0.000000	208.1223

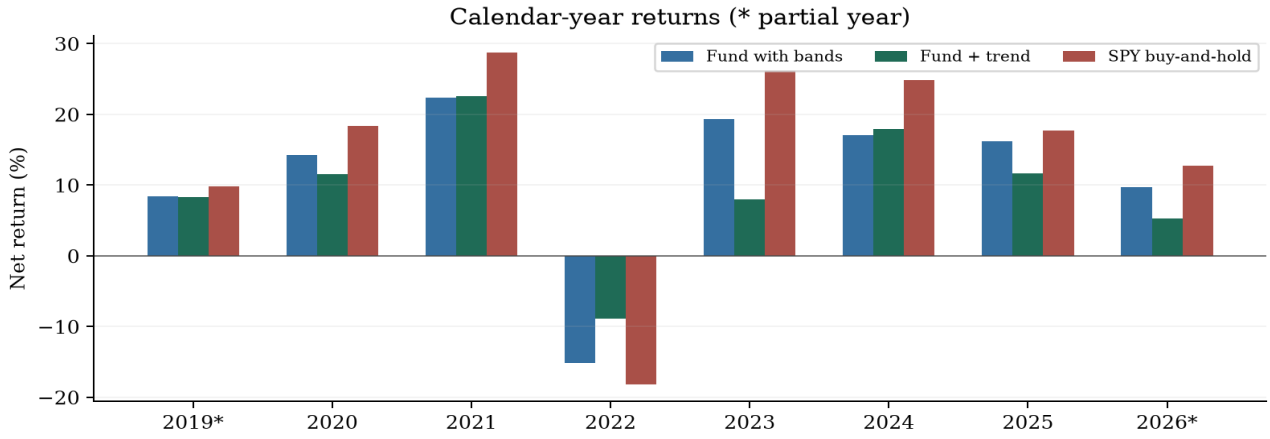


Figure 4. Net calendar-year returns; 2019 and 2026 are partial years.

The saved record reports 17 passed checks. An independent calculation rebuilds positions from trade units and management accrual, then reconciles wealth with prices, cash interest and costs. Maximum daily accounting error is 8.88e-16 initial-capital units. Changing future data or truncating the history must leave earlier wealth unchanged. Artificial software-test fixtures do not enter the research inputs.

With Python 3.13, install requirements.txt and run python run.py. Use python run.py --check for accounting and temporal checks; python -m unittest discover -s tests -v for mechanism tests; and python -m pipeline.qa_artifacts for document checks. The README gives setup instructions and file locations. A clean-folder reproduction is recorded in output/clean_reproduction.json.

The input hashes preserve copy integrity, not vendor certification. The sample includes 2020 and 2022; it does not test the 2001 or 2008 crises. No live orders or Canvas submission have been made.

References

[1] Northwestern University. Module 7: Programming Assignment 3. Revised October 24, 2025. Local snapshot: resources/assignment.md. https://canvas.northwestern.edu/courses/260285/assignments/1818439

[2] Faber, Meb. 2017. A Quantitative Approach to Tactical Asset Allocation, Episode 86, December 13. https://mebfaber.com/2017/12/13/episode-86-quantitative-approach-tactical-asset-allocation/ (checked November 8, 2026). Source for the ten-month rule. This study executes after the signal rather than at the same close.

[3] Clenow, Andreas F. 2019. Trading Evolved: Anyone Can Build Killer Trading Strategies in Python. Independently Published. ISBN 978-1091983786. https://www.followingthetrend.com/trading-evolved/ (author overview checked November 8, 2026). Assignment guide; no exact chapter replication is claimed.

[4] Boyd, Stephen, et al. 2017. Multi-Period Trading via Convex Optimization. Foundations and Trends in Optimization 3(1): 1-76. https://web.stanford.edu/~boyd/papers/cvx_portfolio.html (checked November 8, 2026). Reference for linking allocation, risk and trading costs.

[5] Ledoit, Olivier, and Michael Wolf. 2004. A Well-Conditioned Estimator for Large-Dimensional Covariance Matrices. Journal of Multivariate Analysis 88(2): 365-411. Implemented with sklearn.covariance.LedoitWolf.

[6] Board of Governors of the Federal Reserve System. Three-Month Treasury Constant-Maturity Yield [DGS3MO]. FRED. https://fred.stlouisfed.org/series/DGS3MO (definition checked November 8, 2026). An annualized yield, not a traded total-return series.

[7] Local MSDS 451 Modules 3-6: Programming Assignments 1-2 and Checkpoints A-B. References for fund purpose, catalogue, estimator, constraints, inputs and report style.

[8] Yahoo Finance snapshots preserved in Module 6. Source URLs, retrieval dates and hashes: data/sources.json, data/liquidity_sources.json and data/provenance.json.

Relation to the Module 7 discussion

The discussion used a simpler minimum-variance optimizer, diagonal covariance shrinkage and exact monthly targets. This assignment restores the Checkpoint B mean/covariance estimator and net-Sharpe objective, adds CVaR, bands and fund expenses, and lags cash yields from the interval start. The hypothesis stays the same; the results need not match. Earlier deliverables and the discussion remain unchanged.

Appendix D. Candidate coverage (1/2)

First eligible is the first month passing the history, liquidity and wrapper rules. Maximum target is the largest structural weight. Candidates that never qualify remain visible.

ETF	Quote	First eligible signal	Eligible months	Max target
VOO	USD	2019-06-30	87	0.00%
IVV	USD	Never	0	0.00%
SPY	USD	Never	0	0.00%
VTI	USD	2019-06-30	87	0.00%
ITOT	USD	Never	0	0.00%
VT	USD	2019-06-30	87	0.00%
QQQ	USD	2019-06-30	87	20.00%
VUG	USD	2019-06-30	87	5.83%
VTV	USD	2019-06-30	87	0.00%
VB	USD	2019-06-30	87	0.00%
VYM	USD	2019-06-30	87	0.82%
SCHD	USD	2019-06-30	87	9.53%
VIG	USD	2019-06-30	87	18.04%
DGRO	USD	2019-06-30	87	14.95%
DVY	USD	2019-06-30	87	17.36%
QUAL	USD	2019-06-30	87	0.00%
SPHQ	USD	2019-06-30	87	10.09%
JQUA	USD	2022-11-30	46	0.00%
VOOG	USD	2019-06-30	87	4.49%
VXUS	USD	2019-06-30	87	0.00%
VEA	USD	2019-06-30	87	0.00%
VWO	USD	2019-06-30	87	20.00%
IEMG	USD	2019-06-30	87	1.40%
VBR	USD	2019-06-30	87	0.00%
AVUV	USD	2024-09-30	24	0.00%
MTUM	USD	2019-06-30	87	20.00%
USMV	USD	2019-06-30	87	20.00%
VWRA	USD	Never	0	0.00%
VWRD	USD	2019-06-30	87	2.22%
IWDA	USD	2019-06-30	87	0.00%
VHVE	USD	2025-03-31	18	0.00%

Appendix D. Candidate coverage (2/2)

First eligible is the first month passing the history, liquidity and wrapper rules. Maximum target is the largest structural weight. Candidates that never qualify remain visible.

ETF	Quote	First eligible signal	Eligible months	Max target
ISAC	USD	2020-03-31	78	0.00%
EIMI	USD	2019-06-30	87	13.05%
CSPX	USD	Never	0	0.00%
VUAA	USD	Never	0	0.00%
VUSA	GBP	Never	0	0.00%
CNDX	USD	Never	0	0.00%
IWQU	USD	2020-03-31	77	19.12%
VHYL	GBP	2020-05-31	60	15.55%
VHYA	USD	2026-01-31	8	0.00%
WSML	USD	2024-03-31	30	0.00%
IWMO	USD	2020-06-30	45	9.53%
ZPRV	EUR	2024-12-31	17	0.00%
AVWS	EUR	Never	0	0.00%
BOVA11	BRL	2019-06-30	87	0.00%
BOVV11	BRL	Never	0	0.00%
BBOV11	BRL	Never	0	0.00%
PIBB11	BRL	2019-06-30	22	0.41%
BRAX11	BRL	Never	0	0.00%
SMAL11	BRL	2019-06-30	87	2.63%
DIVO11	BRL	2025-01-31	20	12.35%
BBSD11	BRL	Never	0	0.00%
AUVP11	BRL	Never	0	0.00%
BEST11	BRL	Never	0	0.00%
GOVE11	BRL	Never	0	0.00%
FIND11	BRL	2021-10-31	18	0.00%
IVVB11	BRL	Never	0	0.00%
NASD11	BRL	Never	0	0.00%
WRLD11	BRL	Never	0	0.00%
ACWI11	BRL	Never	0	0.00%
VWRA11	BRL	Never	0	0.00%